

Uzair Moazzam

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EDUCATION

Lahore Garrison University (LGU)

Bachelor of Science in Computer Science — CGPA: 3.18/4.0

Lahore, Pakistan

Sep 2022 – Jun 2026

- Relevant Coursework: Deep Learning, Machine Learning, NLP, Artificial Intelligence, Data Science, Software Engineering, Information Security, Distributed Systems, Computer Networking, Algorithms

WORK EXPERIENCE

Binary Brains

AI/ML Engineer Intern

On-site

Nov 2025 – Feb 2026

- Built and deployed AI/ML pipelines using PyTorch and Hugging Face Transformers across 3 client projects, accelerating each delivery from raw data to a production model.
- Tracked 50+ experiments via Weights & Biases across 3 projects, cutting model selection time by consolidating NLP and computer vision results into a single dashboard.
- Developed an Urdu Fake News Detection system combining DistilBERT embeddings with classical ML classifiers, achieving **83.33% accuracy** on 5,000 articles; applied SMOTE oversampling to raise minority-class recall by **65 percentage points**.

Techtomyy

MERN Stack Developer Intern

Remote

Jul 2025 – Sep 2025

- Built an e-commerce platform (React, Node.js, MongoDB) with cart and checkout flows, a real-time chat app using WebSockets, and an event management system with role-based access control.
- Developed geolocation and routing features for Road Link using the Google Maps API across 2 client deployments, and built a charity platform with secure donation flows and JWT authentication.
- Shipped all 5 systems independently from design to deployment within a 3-month internship.

PROJECTS

SignVerse — Bidirectional Sign Language Recognition System | [GitHub](#) | Final Year Project

Sep 2025 – May 2026

- Built a bidirectional ASL system trained on How2Sign and Google ASL datasets (~16,000-sign vocabulary) with 3 modules: Sign-to-Text, Text-to-Sign avatar animation, and an interactive webcam-based learning module with a 3-tier curriculum.
- Sign-to-Text used a Temporal 1D Transformer over skeletal data; Text-to-Sign ran a 7-stage pipeline (BERT encoder-decoder, PoseFormer, BoneTransformer) to render anatomically realistic avatar output.
- Deployed the backend with FastAPI and OpenCV; integrated Firebase Firestore for session-persistent progress tracking across authenticated and guest users.

Tea Leaf Disease Detection via Swin Transformer & Grad-CAM | [GitHub](#)

2026

- Fine-tuned Swin-T on TeaLeafNet (5,000 images, 4 classes) via 2-phase transfer learning, reaching **98.93%** validation accuracy and **97.87%** test accuracy with TTA×7 at **97 FPS** on Tesla P100.
- Applied Grad-CAM with morphological isolation to confirm disease-specific tissue localisation across all 4 classes, providing explainability validation for clinical use.

RESEARCH & PUBLICATIONS

Fake News Detection in Low-Resource Urdu via Hybrid Deep Learning | [GitHub](#) | Under Review

2025

- Co-authored with LGU and SMU (Dallas, TX) faculty; proposed a 3-tier framework progressing from a Naive Bayes baseline to a weighted stacking ensemble. Best model achieved **83.33% accuracy** and macro F1 of **0.8185**, outperforming the baseline by 13.33 pp on 5,000 verified Urdu articles.

Automated Tea Leaf Disease Detection Using Swin Transformer on TeaLeafNet | [GitHub](#) | Under Review

2026

- Established a reproducible benchmark of TeaLeafNet using Swin-T with full hyperparameter disclosure and Grad-CAM validation, achieving **97.87% test accuracy** at 97 FPS.

TECHNICAL SKILLS

Languages: Python, JavaScript, C, C++

AI / ML: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, NumPy, Pandas, BERT/DistilBERT, Swin Transformer, CNN, XGBoost, SMOTE, Transfer Learning, Fine-Tuning (SFT/PEFT), RAG, FAISS, LangChain, LangGraph, Grad-CAM, Prompt Engineering, OpenCV

Full-Stack: React.js, Node.js, Express.js, MongoDB, MySQL, REST APIs, WebSockets, JWT Auth, Firebase, TailwindCSS, Bootstrap

Tools: Git, GitHub, FastAPI, Weights & Biases, Docker, Jupyter Notebook, Google Colab, VS Code, Kaggle, Overleaf